

Project Design Phase-II Technology Stack (Architecture & Stack)

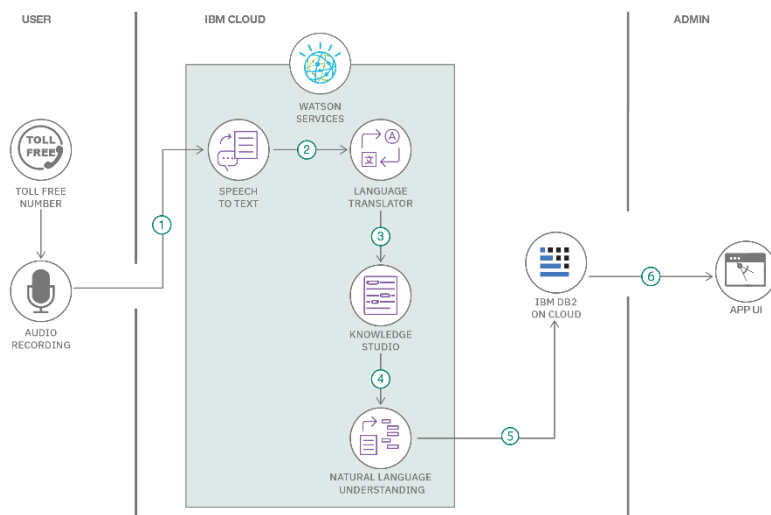
Date	4-07-2026
Team ID	
Project Name	Food Ordering Behaviour and Consumer Trends Analysis Using Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

S.No	Component	Description	Technology
1	User Interface	Interactive dashboard for viewing food ordering trends, customer behaviour, and sales insights	Tableau Dashboard
2	Application Logic-1	Data collection and preprocessing	Python, Pandas
3	Application Logic-2	Data cleaning, transformation, and analysis	Python, NumPy
4	Application Logic-3	Data visualization and dashboard creation	Tableau Desktop
5	Database	Stores food ordering dataset	MySQL / CSV Dataset
6	Cloud Database	Stores backup dataset on cloud (optional)	Google Drive / AWS S3
7	File Storage	Stores CSV files, reports, and Tableau workbook	Local Storage
8	External API-1	Dataset source for food ordering analysis	Kaggle Dataset
9	External API-2	Location-based analysis (optional)	Google Maps API
10	Machine Learning Model	Predict customer ordering trends (optional)	Scikit-learn
11	Infrastructure (Server / Cloud)	System used to develop and deploy the project	Windows PC, Tableau Desktop

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Libraries used for data preprocessing and analysis	Python, Pandas, NumPy
2	Security Implementations	Secure data storage, user authentication, and data validation	MySQL Authentication, File Permissions

S.No	Characteristics	Description	Technology
3	Scalable Architecture	Supports large datasets and dashboard expansion	Tableau, MySQL
4	Availability	Dashboard accessible through Tableau Desktop or Tableau Public	Tableau Desktop, Tableau Public
5	Performance	Fast filtering, visualization, and report generation	Tableau, Optimized SQL Queries